🚀 Lecture 02.07: Relational Operators & Comparison Mechanics

📌 Executive Architectural Summary

Relational (comparison) operators evaluate the order and equality relations between operands according to mathematical order theory, returning a strict Boolean value (`True` or `False`).

🔬 Core Engineering Mechanics

1. Relational Operators Suite

- `==` : Equality ($a = b$)
- `!=` : Inequality (aeb)
- `>` : Strictly greater than ($a > b$)
- `<` : Strictly less than ($a < b$)
- `>=` : Greater than or equal to ($a \geq b$)
- `<=` : Less than or equal to ($a \leq b$)

2. Chained Comparisons Feature

Python uniquely supports chained relational expressions:

$$a < b < c \iff (a < b) \land (b < c)$$

Importantly, operand b is evaluated **only once** in Python's AST execution.

💻 Production Python Implementation

```
x, y = 15, 20

print(f"x == y: {x == y}")
print(f"x != y: {x != y}")
print(f"x > y:  {x > y}")
print(f"x < y:  {x < y}")
print(f"x >= y: {x >= y}")
```

```
print(f"x <= y: {x <= y}")

# Chained Comparison
val = 15
print("Chained: 10 < val <= 20 ->", 10 < val <= 20)
print("Chained: 10 < val < 15 ->", 10 < val < 15)
```